CS105 (DIC on Discrete Structures)
Problem set 7

- Attempt *all* questions.
- Apart from things proved in lecture, you cannot assume anything as "obvious". Either quote previously proved results or provide clear justification for each statement.
- Note that questions on lattices (Qns 1, 2, 5 below) are not in syllabus for the midsem exam.

Part 1

1. Give an example of a poset with five elements that is a lattice and an example of another poset with five elements which is not a lattice.
2. Prove or disprove: Every totally ordered set is a lattice.
3. Give an example of a poset that has 3 maximal elements and 2 minimal elements. Can it have a maximum element?
4. Prove or disprove: the intersection between a chain and an antichain can have at most one element.

Part 2

5. Let L be a lattice. For any two elements $x, y \in L$ we use $x \vee y$ to denote the least upper bound of $\{x, y\}$ and $x \wedge y$ to denote the greatest lower bound of $\{x, y\}$ (note that both these elements exist by the definition of a lattice).
   (a) Show the following properties for all $x, y, z \in L$,
      i. (commutative laws) $x \vee y = y \vee x$ and $x \wedge y = y \wedge x$
      ii. (associative laws) $((x \vee y) \vee z) = (x \vee (y \vee z))$ and $((x \wedge y) \wedge z) = (x \wedge (y \wedge z))$.

iii. (absorption laws) $x \vee (x \wedge y) = x$ and $x \wedge (x \vee y) = x$

iv. (idempotency laws) $x \vee x = x$ and $x \wedge x = x$.

(b) Use the above to prove that every finite nonempty subset of a lattice must have a greatest lower bound and a least upper bound.

6. For all $t > 0$, prove formally that any poset with n elements must have either a chain of length greater than t or an antichain with at least $\frac{n}{t}$ elements.

7. *Consider a permutation of the numbers from 1 to n arranged as a sequence from left to right on a line. Using Mirsky's theorem done in class, prove that there exists a $\sqrt{n}$-length subsequence of these numbers that is completely increasing or completely decreasing as you move from right to left.

   For example, the sequence 2, 3, 4, 7, 9, 5, 6, 1, 8 has an increasing subsequence of length 3, for example: 2, 3, 4, and a decreasing subsequence of length 3, for example: 9, 6, 1. (Hint: Use the previous question!)